

Proliferation and neurogenesis analysis

For proliferation analysis using Ki-67, NPCs were seeded into 24-well plates. The cells were cultured for 7 days, representing approximately three NPC generations. On day 7, cells were collected, and $\sim 1 \times 10^6$ cells were stained with Ki-67-FITC (#130-117-803, Miltenyi Biotec) using the Foxp3/Transcription Factor Staining Buffer Set (#00-5523, Invitrogen), following the manufacturer's protocol.

To evaluate neurogenesis and gliogenesis, NPCs were spontaneously differentiated into human cortical neurons and glial cells. Briefly, 1×10^6 cells were seeded in GelTrex-coated (1:5) 6-well plates and cultured in complete neuronal media containing BrainPhys™ Neuronal Medium, Glutamax (100X), Sodium Pyruvate (100 mM), B-27 (-RA) supplement (50X), N2 (100X), Anti-Anti (100X), Natural Mouse Laminin (1 mg/ml), dbcAMP (500 mg/ml), L-Ascorbic Acid (200 μ M), BDNF (20 μ g/ml), and GDNF (20 μ g/ml). Media was refreshed every three days. On day 25, cells were collected and stained for FACS analysis using surface markers previously described (Yuan, S. H. *et al.* 2011) to differentiate NPCs (CD184+/CD44-/CD24+), neurons (CD184-/CD44-/CD24+), and glia (CD184+/CD44+). CD271, a marker for mesenchymal stem cells, was excluded from the original panel as NPCs were pre-purified via FACS using CD133+/CD184+/CD271- markers prior to differentiation. A minimum of 50,000 cells per gate were acquired using a BD LSRFortessa™ Cell Analyzer at the Yale Flow Cytometry Core. Flow cytometry data were analyzed using FlowJo™ v10.10 Software (BD Life Sciences).

All statistical analyses for flow cytometry assessment were conducted using GraphPad Prism version 9.5.1 (528) for macOS (GraphPad Software, San Diego, CA). Each well was treated as an independent replicate. Differences between knockout (induced) and control (uninduced) groups were assessed by comparing the mean fluorescence intensity (MFI) of the target fluorophore using an unpaired t-test with Welch correction to account for individual group variance. Multiple comparisons were corrected using the False Discovery Rate (FDR) method with a two-stage step-up procedure (Benjamini, Krieger, and Yekutieli) at an FDR threshold of 5%.

Yuan, S. H. *et al.* Cell-surface marker signatures for the isolation of neural stem cells, glia and neurons derived from human pluripotent stem cells. *PLoS One* 6, e17540 (2011).

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